

Lecture Notes on

Numerical Analysis

Prepared By –

Rina Paul

Lecturer, Dept. of Software Engineering,
Metropolitan University, Sylhet

Course Outline

1. Introduction

2. Errors in Numerical Calculation

- Error: Definition, sources, example
- Propagation of errors
- A general error formula – Taylor Series and Reminders

3. Root Finding

- The bisection method
- The iteration method
- The method of false position
- Newton-Raphson Method
- Roots of Polynomial

4. Methods of approximation theory

- Lagrange form
- Divided formula for interpolation

5. Solution of Systems of Linear Equations

- Gaussian Elimination
- The pivoting strategy
- Iteration method solution of tri-diagonal system
- LU decomposition
- Matrix inverse

6. Numerical solution of ODE

- Euler's method (including modified form)
- Runge-Kutta method

7. Numerical Integration

- Trapezoidal method
- Simpson's method
- Weddle's method

8. Eigen value problem for matrices.

1. Introduction

Numerical methods are techniques by which mathematical problems are formulated so that they can be solved with arithmetic operations. Although there are many kinds of numerical methods they have one common characteristic: most of them involve large number of tedious arithmetic calculations. With the development of fast, efficient digital computers, the role of numerical methods in engineering problem solving has increased dramatically in recent years. The great advantage of using numerical analysis is that it investigates and provides accurate solutions to real-life problems from the field of science, engineering, biology, astrophysics and finance.

The word 'analysis' generally means to solve a problem through a set of equations and further reduce these equations using the methodologies of algebra, partial differential equations, calculus and other related fields of mathematics. On similar grounds, numerical analysis implements arithmetic algorithms: addition, subtraction, multiplication, division and comparison to obtain numerical solutions. A computer precisely performs these operations, meaning that numerical analysis and computers are intimately related.

The problem of continuous mathematics generally arises throughout the natural sciences, business management, engineering, astronomy and medicine. It is always more practical to carry out the tedious arithmetic operations using a computer. Before the mid-20th century, until the advent of modern computers, all these repetitive operations had to be performed by manual interpolation. The overall agenda of numerical analysis is to give an *approximate*, but *accurate* solution to the advanced problem. The necessity for accuracy, is, of course, determined by the variety of applications.

Why Numerical Analysis?

- Extremely powerful problem-solving tool: Capable of handling large systems of equations, nonlinearities, and complicated geometries.
- Knowledge of basic theory of numerical methods will help to properly use pre-packaged computer programmes that involve numerical methods.
- Better concept of numerical method might help one to develop his own computer program.
- Efficient vehicle for learning to use computers
- Reinforce the understanding of mathematics

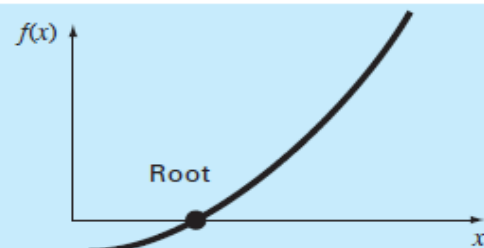
Application –

- Car safety enhancements
- Airflow patterns in the Respiratory Tract
- Onset and Progression of tumour cells
- Financial industry
- Transportation of chemicals in the body

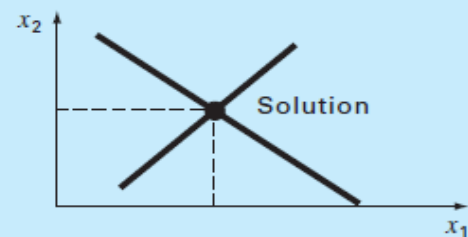
- High hydrostatic pressure (HHP) processing
- Weather predictions
- Spacecraft Dynamics
- Price estimation by airlines
- Machine learning

What we are going to learn?

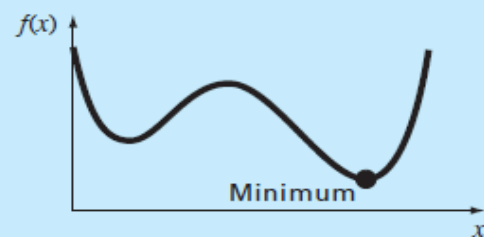
(a) *Part 2: Roots of equations*
Solve $f(x) = 0$ for x .



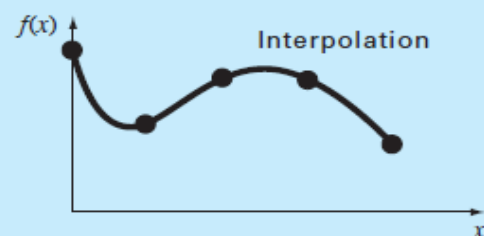
(b) *Part 3: Linear algebraic equations*
Given the a 's and the c 's, solve
 $a_{11}x_1 + a_{12}x_2 = c_1$
 $a_{21}x_1 + a_{22}x_2 = c_2$
 for the x 's.



(c) *Part 4: Optimization*
Determine x that gives optimum $f(x)$.



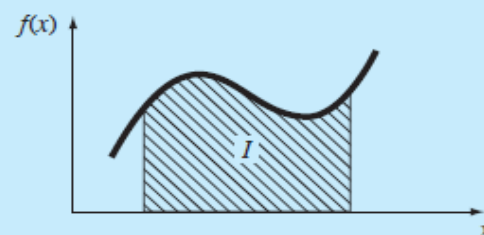
(d) *Part 5: Curve fitting*



(e) *Part 6: Integration*

$$I = \int_a^b f(x) dx$$

Find the area under the curve.



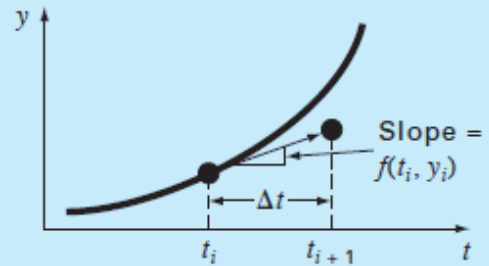
(f) Part 7: Ordinary differential equations

Given

$$\frac{dy}{dt} \approx \frac{\Delta y}{\Delta t} = f(t, y)$$

solve for y as a function of t .

$$y_{i+1} = y_i + f(t_i, y_i) \Delta t$$

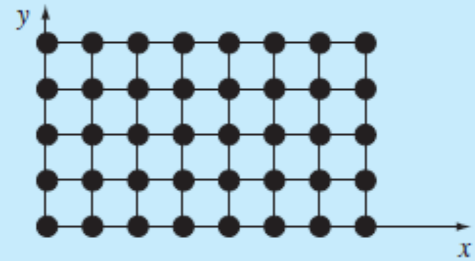


(g) Part 8: Partial differential equations

Given

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$$

solve for u as a function of x and y



2. Errors in Numerical Analysis

Numerical errors arise from the use of approximations to represent exact mathematical operations and quantities. The relationship between the exact, or true result and the approximation can be formulated as

$$\text{True value} = \text{approximation} + \text{error} \quad (2.1)$$

By rearranging Eq. (2.1), we find that the numerical error is equal to the discrepancy between the truth and the approximation, as in

$$E_t = \text{true value} - \text{approximation} \quad (2.2)$$

Where E_t is used to designate the exact value of the error.

Accuracy and Precision –

Accuracy refers to how closely a computed or measured value agrees with the true value.

Precision refers to how closely individual computed or measured values agree with each other.

Significant Figure –

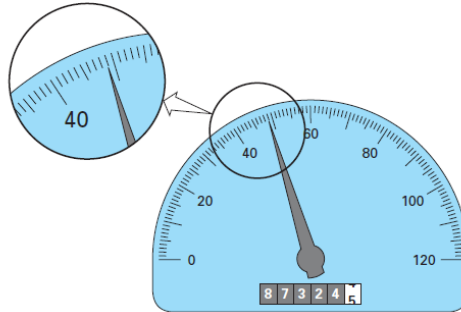


Fig 2.1: An automobile speedometer and odometer illustrating the concept of a significant figure

The concept of a significant figure or digit has been developed to formally designate the reliability of a numerical value. The significant digits of a number are those that can be used with confidence. They correspond to the number of certain digits plus one estimated digit. (e.g. Fig 2.1)

Types of Errors –

- Inherent Errors
- Round-off Errors
- Truncation Errors

Inherent Errors:

- Error due to human mistakes or committed in collection of data. Can be minimized by using better techniques for the collection of data.
- Main Cause - Uncertainty in measurement, non-availability of exact expressions for certain numbers

Round-off Errors:

- Errors due to the limitations of the computing aids, mathematical tables, desk calculators or the digital computers numbers have to be rounded off, causing what are called rounding off errors.
- Can be minimized by using computing aids of higher precision. To reduce round off error there is an useful rule –
“At each step of the computation, retain at least one more significant figure than that given in the data, perform the last operation and finally round off”

Truncation Errors:

- Errors which arise when an infinite process is approximated by finite one. (e.g. when a definite integral is computed by Simpson’s rule or when an infinite series is summed by any approximate formula)

Absolute, Relative and Percentage errors –

The numerical difference between the true value of a quantity and its approximate value is defined as *Absolute Error*.

Let X be the true value of a quantity and X_1 be its approximate value then the absolute error E_A is given by,

$$E_A = X - X_1 = \delta X$$

The ratio of the absolute error and the true value is defined as *Relative Error*.

$$E_R = \frac{E_A}{X} = \frac{\delta X}{X}$$

The 100 times the Relative error is defined as the *percentage error*.

$$E_p = 100 E_R$$

Floating-Point Representation:

Fractional quantities are typically represented in computers using floating-point form. In this approach, the number is expressed as a fractional part, called a mantissa or significand, and an integer part, called an exponent or characteristic, as in

$$m \cdot b^e$$

where m = the mantissa, b = the base of the number system being used, and e = the exponent.

For instance, the number 156.78 could be represented as 0.15678×10^3 in a floating point base-10 system.

The consequence of normalization is that the absolute value of m is limited. That is,

$$\frac{1}{b} \leq m < 1$$

where b = the base. For example, for a base-10 system, m would range between 0.1 and 1, and for a base-2 system, between 0.5 and 1.

Example -

The number 123.45 can be represented as a decimal floating-point number with the integer 12345 as the mantissa and a 10^{-2} power term, also called characteristics. where -2 is the exponent (and 10 is the base). Its value is given by the following arithmetic:

$$123.45 = 12345 \times 10^{-2}.$$

The same value can also be represented in normalized form with 1.2345 as the fractional coefficient, and $+2$ as the exponent (and 10 as the base)

$$123.45 = 1.2345 \times 10^{+2}.$$

Schmid, however, called this representation with a mantissa ranging between 1.0 and 10 a modified normalized form.

Finally, the value can be represented in the format given by the Language Independent Arithmetic standard and several programming language standards, including Ada, C, Fortran and Modula-2, as

$$123.45 = 0.12345 \times 10^{+3}.$$

Schmid called this representation with a mantissa ranging between 0.1 and 1.0 the true normalized form.

Machine Epsilon:

The Interval between Numbers, Δx , Increases as the Numbers Grow in Magnitude. It is this characteristic, of course, that allows floating-point representation to preserve significant digits. However, it also means that quantizing errors will be proportional to the magnitude of the number being represented. For normalized floating-point numbers, this proportionality can be expressed, for cases where chopping is employed, as

$$\frac{|\Delta x|}{|x|} \leq \varepsilon$$

and, for cases where rounding is employed, as

$$\frac{|\Delta x|}{|x|} \leq \frac{\varepsilon}{2}$$

Where ε is referred as machine epsilon which can be computed as $\varepsilon = b^{1-t}$ where b is the number base and t is the number of significant digits in the mantissa.

In brief, Machine epsilon or machine precision gives an upper bound on the relative approximation error due to rounding in floating point arithmetic. Rounding is a procedure for choosing the representation of a real number in a floating point number system. For a number system and a rounding procedure, machine epsilon is the maximum relative error of the chosen rounding procedure.

Mathematical formula of error propagation for function of single variable

Suppose that we have a function $f(x)$ that is dependent on a single independent variable x .

Assume that $\tilde{x}$ is an approximation of x . We, therefore, would like to assess the effect of the discrepancy between x and $\tilde{x}$ on the value of the function. That is, we would like to estimate

$$\Delta f(\tilde{x}) = |f(x) - f(\tilde{x})|$$

The problem with evaluating $\Delta f(\tilde{x})$ is that $f(x)$ is unknown because x is unknown. We can overcome this difficulty if $\tilde{x}$ is close to x and $f(\tilde{x})$ is continuous and differentiable. If these conditions hold, a Taylor series can be employed to compute $f(x)$ near $f(\tilde{x})$, as in

$$f(x) = f(\tilde{x}) + (x - \tilde{x})f'(\tilde{x}) + \frac{(x - \tilde{x})^2}{2!}f''(\tilde{x}) + \dots \dots \dots$$

Dropping the second- and higher-order terms and rearranging yields

$$f(x) - f(\tilde{x}) = (x - \tilde{x})f'(\tilde{x})$$

$$\Rightarrow \Delta f(\tilde{x}) = \Delta \tilde{x} |f'(\tilde{x})| \dots \dots \dots \textbf{(1)}$$

Where $\Delta f(\tilde{x}) = f(x) - f(\tilde{x})$ represents an estimate of the error of the function and $\Delta \tilde{x} = (x - \tilde{x})$ represents an estimate of the error of x .

Problem Statement: Given a value of $\tilde{x} = 2.5$ with an error of $\Delta \tilde{x} = 0.01$, estimate the resulting error in the function, $f(x) = x^3$.

Solution. Using Eq. (1),

$$\Delta f(\tilde{x}) = 3(2.5)^2(0.01) = 0.1875$$

Because $f(2.5) = 15.625$, we predict that

$f(2.5) = 15.625 \pm 0.1875$ or that the true value lies between 15.4375 and 15.8125.

For Function of multiple variables

$$\Delta f(\widetilde{x}_1, \widetilde{x}_2, \dots \dots \widetilde{x}_n) = \left| \frac{\partial f}{\partial x_1} \right| \Delta \widetilde{x}_1 + \left| \frac{\partial f}{\partial x_2} \right| \Delta \widetilde{x}_2 + \dots \dots + \left| \frac{\partial f}{\partial x_n} \right| \Delta \widetilde{x}_n \dots (2)$$

Problem Statement. The deflection y of the top of a sailboat mast is

$$y = \frac{FL^4}{8EI}$$

where F = a uniform side loading (N/m), L = height (m), E = the modulus of elasticity (N/m²), and I = the moment of inertia (m⁴). Estimate the error in y given the following data:

$$\begin{array}{ll} \tilde{F} = 750 \text{ N/m} & \Delta \tilde{F} = 30 \text{ N/m} \\ \tilde{L} = 9 \text{ m} & \Delta \tilde{L} = 0.03 \text{ m} \\ \tilde{E} = 7.5 \times 10^9 \text{ N/m}^2 & \Delta \tilde{E} = 5 \times 10^7 \text{ N/m}^2 \\ \tilde{I} = 0.0005 \text{ m}^4 & \Delta \tilde{I} = 0.000005 \text{ m}^4 \end{array}$$

Solution :

Employing Equation (2) gives,

$$\Delta y(\tilde{F}, \tilde{L}, \tilde{E}, \tilde{I}) = \left| \frac{\partial y}{\partial F} \right| \Delta \tilde{F} + \left| \frac{\partial y}{\partial L} \right| \Delta \tilde{L} + \left| \frac{\partial y}{\partial E} \right| \Delta \tilde{E} + \left| \frac{\partial y}{\partial I} \right| \Delta \tilde{I}$$

or

$$\Delta y(\tilde{F}, \tilde{L}, \tilde{E}, \tilde{I}) \cong \frac{\tilde{L}^4}{8\tilde{E}\tilde{I}} \Delta \tilde{F} + \frac{\tilde{F}\tilde{L}^3}{2\tilde{E}\tilde{I}} \Delta \tilde{L} + \frac{\tilde{F}\tilde{L}^4}{8\tilde{E}^2\tilde{I}} \Delta \tilde{E} + \frac{\tilde{F}\tilde{L}^4}{8\tilde{E}\tilde{I}^2} \Delta \tilde{I}$$

Substituting the appropriate values gives

$$\Delta y = 0.006561 + 0.002187 + 0.001094 + 0.00164 = 0.011482$$

Therefore, $y = 0.164025 \pm 0.011482$. In other words, y is between 0.152543 and 0.175507 m. The validity of these estimates can be verified by substituting the extreme values for the variables into the equation to generate an exact minimum of

$$y_{\min} = \frac{720(8.97)^4}{8(7.55 \times 10^9)0.000505} = 0.152818$$

and

$$y_{\max} = \frac{780(9.03)^4}{8(7.45 \times 10^9)0.000495} = 0.175790$$

Thus, the first-order estimates are reasonably close to the exact values.

Problem:

Use Taylor series expansions with $n = 0$ to 6 to approximate $f(x) = \cos x$ at $x_{i+1} = \frac{\pi}{3}$ on the basis of the value of $f(x)$ and its derivatives at $x_i = \frac{\pi}{4}$.

Solution:

Given $x_{i+1} = \frac{\pi}{3}$ and $x_i = \frac{\pi}{4}$.

Suppose $h = x_{i+1} - x_i = \frac{\pi}{3} - \frac{\pi}{4} = \frac{\pi}{12}$

Taylor series expansion of $f(x)$ with respect to the value of $f(x)$ and its derivatives at $x_i = \frac{\pi}{4}$ is as follow –

$$f(x_{i+1}) = f(x_i) + hf'(x_i) + \frac{h^2}{2!}f''(x_i) + \frac{h^3}{3!}f'''(x_i) + \dots \dots \dots$$

Now, zeroth order approximation –

$$f(x_{i+1}) = f(x_i) + hf'(x_i) + \frac{h^2}{2!}f''(x_i) + \frac{h^3}{3!}f'''(x_i) + \dots \dots \dots$$

$$f(x_{i+1}) = f(x_i) \Rightarrow f\left(\frac{\pi}{3}\right) = f\left(\frac{\pi}{4}\right) = \cos \frac{\pi}{4} = 0.707106781$$

$$\text{Error, } \epsilon_1 = \left| \frac{0.5 - 0.707106781}{0.5} \right| \times 100\% = 41.4\%$$

Then, First order approximation,

$$f(x_{i+1}) = f(x_i) + hf'(x_i) + \frac{h^2}{2!}f''(x_i) + \frac{h^3}{3!}f'''(x_i) + \dots \dots \dots$$

$$f(x_i) = \cos x_i \Rightarrow f'(x_i) = -\sin x_i \Rightarrow f'\left(\frac{\pi}{4}\right) = -\sin \frac{\pi}{4} = -0.707106781$$

$$\text{So, } f(x_{i+1}) = f(x_i) + hf'(x_i) \Rightarrow f\left(\frac{\pi}{3}\right) = f\left(\frac{\pi}{4}\right) + \frac{\pi}{12} \times -0.707106781 = 0.521986659$$

$$\text{Error, } \epsilon_2 = \left| \frac{0.5 - 0.521986659}{0.5} \right| \times 100\% = 4.4\%$$

Second Order Approximation:

$$f(x_{i+1}) = f(x_i) + hf'(x_i) + \frac{h^2}{2!} f''(x_i) + \frac{h^3}{3!} f'''(x_i) + \dots \dots \dots$$

$$f\left(\frac{\pi}{3}\right) = f\left(\frac{\pi}{4}\right) + \frac{\pi}{12} \times f'\left(\frac{\pi}{4}\right) + \frac{\left(\frac{\pi}{12}\right)^2}{2} \times f''\left(\frac{\pi}{4}\right)$$

$$\Rightarrow f\left(\frac{\pi}{3}\right) = 0.707106781 + \frac{\pi}{12} \times \left(-\sin \frac{\pi}{4}\right) - \frac{\left(\frac{\pi}{12}\right)^2}{2} \times \left(\cos \frac{\pi}{4}\right) = 0.497754491$$

$$\text{Error, } \varepsilon_3 = \left| \frac{0.5 - 0.497754491}{0.5} \right| \times 100\% = 0.449\%$$

Similarly we do the rest of the steps $n = 3$ to 6

Order n	$f^{(n)}(x)$	$f(\pi/3)$	ε_f
0	$\cos x$	0.707106781	-41.4
1	$-\sin x$	0.521986659	-4.4
2	$-\cos x$	0.497754491	0.449
3	$\sin x$	0.499869147	2.62×10^{-2}
4	$\cos x$	0.500007551	-1.51×10^{-3}
5	$-\sin x$	0.500000304	-6.08×10^{-5}
6	$-\cos x$	0.499999988	2.44×10^{-6}

Which means we have attained $100 - (2.44 \times 10^{-6}) = 99.999998\%$ of the true value.

Exercises

- [1] What do you mean by Truncation error? Explain with examples
- [2] What do you understand by Machine Epsilon of a computer? Explain.
- [3] What is round off error?
- [4] Round –off the numbers 965.25278 and 37.4623553 to five significant figures and compute e_p .
- [5] Find the percentage error if 625.483 is approximated to three significant figures.
- [6] Round –off the following numbers to four significant figures.
3.65784, 4685928, 0.70035, 0.00045637, 30.0567, 0.4672389.
- [7] If $X=2.536$, find absolute and relative error when
 - (i) X is truncated to two decimal digits
 - (ii) X is rounded-off
- [8] Find the approximate value of $1/3$ such that it contains 0.30, 0.33 & 0.34
- [9] Compute relative and percentage error when 100500 is approximated by 100000
- [10] Round –off the numbers 437.261 to four significant figures and compute e_p .

3. Root Finding Methods

3.1 Bisection Method

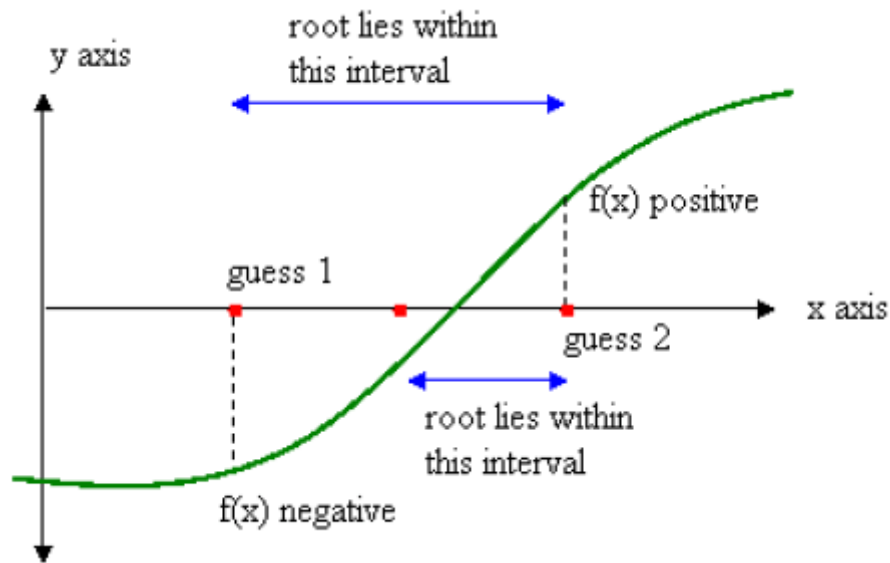


Figure: Bisection method concept

Algorithm –

Bisection method Steps	
Step-1:	Find points a and b such that $a < b$ and $f(a) \cdot f(b) < 0$.
Step-2:	Take the interval $[a, b]$ and find next value $t = (a + b)/2$
Step-3:	If $f(t) = 0$ then t is an exact root, stop. else if $f(a) \cdot f(t) < 0$ then $b = t$, else if $f(t) \cdot f(b) < 0$ then $a = t$.
Step-4:	Repeat steps 2 & 3 until $f(t) = 0$ or $ f(t) \leq \text{Accuracy}$

Example-1

Find a root of an equation $f(x) = x^3 - x - 1$ correct up to three decimal points using Bisection method.

Solution:

Here we are to solve, $x^3 - x - 1 = 0$

Step 1:

Here

x	0	1	2
$f(x)$	-1	-1	5

The diagram shows a table with two rows. The first row is labeled x and contains values 0, 1, and 2. The second row is labeled $f(x)$ and contains values -1, -1, and 5. Blue arrows point from labels 'a' and 'b' to the cell containing '1' in the first row. Another blue arrow points from 'f(a)' to the cell containing '-1' in the second row. A final blue arrow points from 'f(b)' to the cell containing '5' in the second row.

[a and b are chosen in such a way that $f(a) \cdot f(b) < 0$]

Step 2:

Suppose, $t = \frac{a+b}{2} = 1.5$

$f(t) = f(1.5) = 0.875 > 0$

Step 3 : Since $f(t) \cdot f(a) < 0 \Rightarrow b = t$;

So, now, $a = 1, b = 1.5$

Repeating the steps again and again we get the following table –

<i>n</i>	<i>a</i>	<i>f(a)</i>	<i>b</i>	<i>f(b)</i>	<i>c = (a + b)/2</i>	<i>f(c)</i>	<i>Update</i>
1	1	−1	2	5	1.5	0.875	<i>b = c</i>
2	1	−1	1.5	0.875	1.25	−0.29688	<i>a = c</i>
3	1.25	−0.29688	1.5	0.875	1.375	0.22461	<i>b = c</i>
4	1.25	−0.29688	1.375	0.22461	1.3125	−0.05151	<i>a = c</i>
5	1.3125	−0.05151	1.375	0.22461	1.34375	0.08261	<i>b = c</i>
6	1.3125	−0.05151	1.34375	0.08261	1.32812	0.01458	<i>b = c</i>
7	1.3125	−0.05151	1.32812	0.01458	1.32031	−0.01871	<i>a = c</i>
8	1.32031	−0.01871	1.32812	0.01458	1.32422	−0.00213	<i>a = c</i>
9	1.32422	−0.00213	1.32812	0.01458	1.32617	0.00621	<i>b = c</i>
10	1.32422	−0.00213	1.32617	0.00621	1.3252	0.00204	<i>b = c</i>
11	1.32422	−0.00213	1.3252	0.00204	1.32471	−0.00005	<i>a = c</i>

So, approximate root of the equation $x^3 - x - 1 = 0$ using Bisection mehtod is 1.32471.

3.2 False Position Method (Regula-Falsi Method)

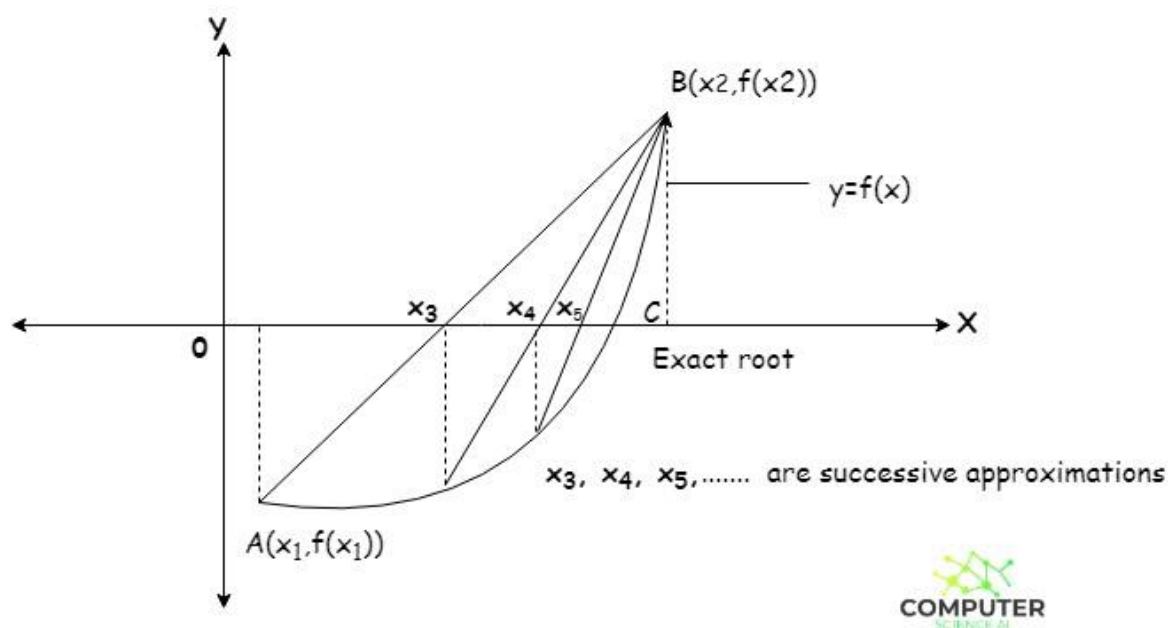


Fig. False Position / Regula-Falsi Method

Figure – False Position Method

Consider we are to find a real root of $y = f(x)$. We must choose two points x_1 and x_2 ($x_1 < x_2$) first such that, $f(x_1) \cdot f(x_2) < 0$

Equation of the line going through $(x_1, f(x_1))$ and $(x_2, f(x_2))$ is –

$$\frac{x-x_1}{x_1-x_2} = \frac{y-f(x_1)}{f(x_1)-f(x_2)} \quad \dots\dots (1)$$

$$\Rightarrow \frac{y-f(x_1)}{x-x_1} = \frac{f(x_1)-f(x_2)}{x_1-x_2}$$

$$\Rightarrow \frac{y-f(x_1)}{x-x_1} = \frac{f(x_2)-f(x_1)}{x_2-x_1} \quad \dots\dots (2)$$

Suppose, $y = 0$, when $x = x_3$, then (2) reduces to,

$$\frac{0-f(x_1)}{x_3-x_1} = \frac{f(x_2)-f(x_1)}{x_2-x_1}$$

$$\Rightarrow -\frac{f(x_1)}{f(x_2)-f(x_1)}(x_2-x_1) = x_3-x_1$$

$$\Rightarrow x_3 = x_1 - \frac{f(x_1)}{f(x_2)-f(x_1)}(x_2-x_1) \quad \text{[First approximation]}$$

Now, check if $f(x_3) = 0$, else do the following step.

(i) If $f(x_1) \cdot f(x_3) < 0$ (i.e. have different signs) then $x_2 = x_3$ (x_2 will be updated by x_3)

(ii) Or if $f(x_2) \cdot f(x_3) < 0$ (i.e. have different signs) then $x_1 = x_3$ (x_1 will be updated by x_3)

Steps in Table –

False Position method (regula falsi method) Steps (Rule)	
Step-1:	Find points x_1 and x_2 such that $x_1 < x_2$ and $f(x_1) \cdot f(x_2) < 0$.
Step-2:	Take the interval $[x_1, x_2]$ and find next value using Formula: $x_3 = x_1 - \frac{f(x_1)}{f(x_2) - f(x_1)}(x_2 - x_1)$
Step-3:	If $f(x_3) = 0$ then x_3 is an exact root, else if $f(x_1) \cdot f(x_3) < 0$ then $x_2 = x_3$, else if $f(x_2) \cdot f(x_3) < 0$ then $x_1 = x_3$.
Step-4:	Repeat steps 2 & 3 until $f(x_3) = 0$ or $ f(x_3) \leq \text{Accuracy}$

Example - 2

Find a root of an equation $f(x) = 2x^3 - 2x - 5$ using False Position method (regula falsi method)

Solution:

Here $2x^3 - 2x - 5 = 0$

Let $f(x) = 2x^3 - 2x - 5$

Here

x	0	1	2
$f(x)$	-5	-5	7

So, We have $x_1 = 1, f(x_1) = -5$ and $x_2 = 2, f(x_2) = 7$

Such that $f(x_1) \cdot f(x_2) = -10 < 0$

So, There must be a root within the interval $[1, 2]$.

First Iteration:

$$x_3 = x_1 - \frac{f(x_1)}{f(x_2) - f(x_1)}(x_2 - x_1)$$

$$\Rightarrow x_3 = 1 - \frac{-5}{7+5}(2 - 1) = 1 + \frac{5}{12} = \frac{17}{12} = 1.41667$$

Now, $f(x_3) = f(1.41667) = 2 \cdot 1.41667^3 - 2 \cdot 1.41667 - 5 = -2.14699 < 0$

Since $f(x_3) \cdot f(x_2) < 0 \Rightarrow x_1 = x_3$ (So, value of x_1 will be updated by x_3)

And other steps will happen iteratively since the required accuracy is attained.

All the iterations are given in the following table –

n	x_1	$f(x_1)$	x_2	$f(x_2)$	x_3	$f(x_3)$	Update
1	1	-5	2	7	1.41667	-2.14699	$x_1 = x_3$
2	1.41667	-2.14699	2	7	1.55359	-0.60759	$x_1 = x_3$
3	1.55359	-0.60759	2	7	1.58924	-0.15063	$x_1 = x_3$
4	1.58924	-0.15063	2	7	1.59789	-0.0361	$x_1 = x_3$
5	1.59789	-0.0361	2	7	1.59996	-0.00858	$x_1 = x_3$
6	1.59996	-0.00858	2	7	1.60045	-0.00203	$x_1 = x_3$
7	1.60045	-0.00203	2	7	1.60056	-0.00048	$x_1 = x_3$

Therefore, approximate root of the equation $2x^3 - 2x - 5 = 0$ using False Position method is 1.60056.

(Solved)

3.3 Newton-Raphson Method

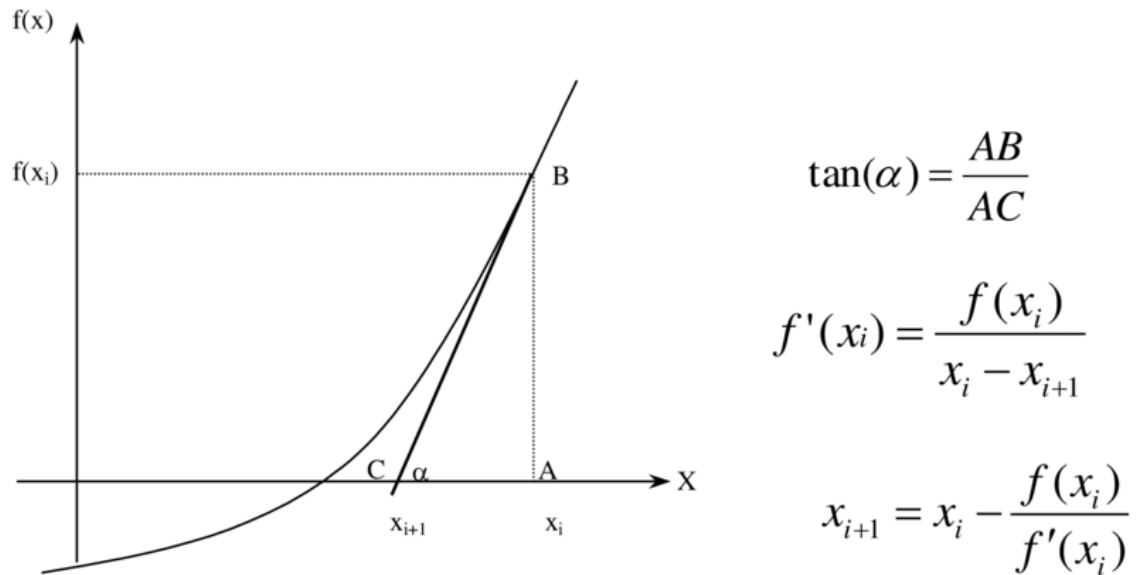


Figure – Newton Raphson Method

Newton Raphson method Steps (Rule)	
Step-1:	Find points a and b such that $a < b$ and $f(a) \cdot f(b) < 0$.
Step-2:	Take the interval $[a, b]$ and find next value $x_0 = \frac{a+b}{2}$
Step-3:	Find $f(x_0)$ and $f'(x_0)$ $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$
Step-4:	If $f(x_1) = 0$ then x_1 is an exact root, else $x_0 = x_1$
Step-5:	Repeat steps 3 and 4 until $f(x_i) = 0$ or $ f(x_i) \leq \text{Accuracy}$

Example-3

Find a root of an equation $f(x) = x^3 - x - 1$ using Newton Raphson method

Solution:

Here $x^3 - x - 1 = 0$

Let $f(x) = x^3 - x - 1$

$$\therefore f'(x) = 3x^2 - 1$$

Here

x	0	1	2
$f(x)$	-1	-1	5

Here $f(1) = -1 < 0$ and $f(2) = 5 > 0$

$\therefore$ Root lies between 1 and 2

$$x_0 = \frac{1 + 2}{2} = 1.5$$

1st iteration :

$$f(x_0) = f(1.5) = 0.875$$

$$f'(x_0) = f'(1.5) = 5.75$$

$$\therefore x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \Rightarrow x_1 = 1.5 - \frac{0.875}{5.75} = 1.34783$$

Now, Putting $x_0 = x_1$ (Updating x_0 by the value of x_1), we'll repeat the steps again until we attain desired accuracy.

n	x_0	$f(x_0)$	$f'(x_0)$	x_1	<i>Update</i>
1	1.5	0.875	5.75	1.34783	$x_0 = x_1$
2	1.34783	0.10068	4.44991	1.3252	$x_0 = x_1$
3	1.3252	0.00206	4.26847	1.32472	$x_0 = x_1$
4	1.32472	0	4.26463	1.32472	$x_0 = x_1$

So, approximate root of the equation $x^3 - x - 1 = 0$ using Newton Raphson method is 1.32472. (We have obtained the root way more faster using this method)

(Solved)

3.4 Fixed Point Iteration Method

Fixed Point Iteration method Steps (Rule)	
Step-1:	First write the equation $f(x) = 0$ in the form $x = \phi(x)$
Step-2:	Find points a and b such that $a < b$ and $f(a) \cdot f(b) < 0$.
Step-3:	If $f(a)$ is more closer to 0 then $f(b)$ then $x_0 = a$ else $x_0 = b$
Step-4:	$x_1 = \phi(x_0)$ If $f(x_1) = 0$, then x_1 is the exact root else $x_0 = x_1$ (<i>Updated x_0</i>) Repeat until $ x_1 - x_0 \approx 0$

Example - 4

Find a root of an equation $f(x) = 2x^3 - 2x - 5$ using Fixed point iteration method.

Solution:

Let $f(x) = 2x^3 - 2x - 5$

$$2x^3 - 2x - 5 = 0$$

$$2x^3 = 2x + 5 \Rightarrow x^3 = \frac{2x + 5}{2} \Rightarrow x = \sqrt[3]{\frac{2x + 5}{2}}$$

Thus, $\phi(x) = \sqrt[3]{\frac{2x+5}{2}}$

Here

x	0	1	2
$f(x)$	-5	-5	7

Here $f(1) = -5 < 0$ and $f(2) = 7 > 0$

$\therefore$ Root lies between 1 and 2

$$x_0 = \frac{1+2}{2} = 1.5$$

$$x_1 = \phi(x_0) = \phi(1.5) = 1.5874$$

$$f(x_1) = -0.174815 \text{ and}$$

So, $x_0 = x_1$ and let's follow the steps until $|x_1 - x_0| \leq 10^{-3}$. All the steps are written in the following table.

n	x_0	$x_1 = \phi(x_0)$	<i>Update</i>	<i>Difference</i> $x_1 - x_0$
2	1.5	1.5874	$x_0 = x_1$	0.0874
3	1.5874	1.59888	$x_0 = x_1$	0.01148
4	1.59888	1.60037	$x_0 = x_1$	0.0015
5	1.60037	1.60057	$x_0 = x_1$	0.00019

Approximate root of the equation $2x^3 - 2x - 5 = 0$ using Iteration method is 1.60057.

(Solved)

Exercises –

Newton-Raphson Method

- [1] Use Newton-Raphson Method to obtain a root of the equation $x^3 - 3x = 5$ correct to four decimal places.
- [2] Use Newton-Raphson Method to obtain a root of the equation $x^3 = 6x - 4$, correct to five decimal places.
- [3] Use Newton-Raphson Method to obtain a root of the equation $2x^3 - 3x = 5$ correct to five decimal places.

Bisection Method

- [4] Use Bisection Method to obtain a root of the equation $x^3 + x^2 = 1$, correct to three decimal places.
- [5] Use Bisection Method to obtain a root of the equation $x^3 - x = 1$ Correct to three decimal places.

False-Position Method

- [6] Use False-Position Method to obtain a root of the equation $x^3 = 4x - 1$ correct to four decimal places.
- [7] Use False-Position Method to obtain a root of the equation $x^3 = 2x + 5$, correct four decimal places.

Method of Iteration

- [8] Find a real root of the equation, $x^3 = 2x + 5$ by the method of Iteration, correct to four decimal places.

4. *Interpolation*

Let the set of tabular values $(x_0, y_0), (x_1, y_1), \dots \dots (x_n, y_n)$ satisfy the relation $y = f(x)$ where the explicit nature of $f(x)$ is not known, say $\varphi(x)$, such that $f(x)$ and $\varphi(x)$ agree at the set of tabulated points. Such a process is called interpolation.

If $\varphi(x)$ is a polynomial, then the process is called polynomial interpolation and $\varphi(x)$ is called interpolating polynomial.

- Interpolation with equally spaced points
 - Newton's Forward difference formula
 - Newton's Backward difference formula
 - Newton's Central Difference formula
- Interpolation with unevenly spaced points
 - Lagrange's interpolation formula
 - Newton's general interpolation formula with divided difference

4.1 Newton's forward difference formula

Formula

Newton's Forward Difference formula

$$p = \frac{x - x_0}{h}$$

$$y(x) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!} \cdot \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \cdot \Delta^3 y_0 + \frac{p(p-1)(p-2)(p-3)}{4!} \cdot \Delta^4 y_0 + \dots$$

Examples

1. Find Solution using Newton's Forward Difference formula

x	$f(x)$
1891	46
1901	66
1911	81
1921	93
1931	101

Find value at $x = 1895$.

Solution:

The value of table for x and y

x	1891	1901	1911	1921	1931
y	46	66	81	93	101

Newton's forward difference interpolation method to find solution

Newton's forward difference table is

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1891	46				
		20			
1901	66		-5		
		15		2	
1911	81		-3		-3
		12		-1	
1921	93		-4		
		8			
1931	101				

The value of x at we want to find the $f(x)$: $x = 1895$

$$h = x_1 - x_0 = 1901 - 1891 = 10$$

$$p = \frac{x - x_0}{h} = \frac{1895 - 1891}{10} = 0.4$$

Newton's forward difference interpolation formula is

$$y(x) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!} \cdot \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \cdot \Delta^3 y_0 + \frac{p(p-1)(p-2)(p-3)}{4!} \cdot \Delta^4 y_0 + \dots$$

$$\Rightarrow y(1895) = 46 + 0.4 \times 20 + \frac{0.4(0.4 - 1)}{2} \times -5 + \frac{0.4(0.4 - 1)(0.4 - 2)}{6} \times 2$$

$$+ \frac{0.4(0.4 - 1)(0.4 - 2)(0.4 - 3)}{24} \times -3$$

$$y(1895) = 46 + 8 + 0.6 + 0.128 + 0.1248$$

$$y(1895) = 54.8528$$

Solution of newton's forward interpolation method $y(1895) = 54.8528$.

(Solved)

4.2 Newton's Backward Difference formula

Formula

Newton's Backward Difference formula

$$p = \frac{x - x_n}{h}$$

$$y(x) = y_n + p \nabla y_n + \frac{p(p+1)}{2!} \cdot \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \cdot \nabla^3 y_n$$

$$+ \frac{p(p+1)(p+2)(p+3)}{4!} \cdot \nabla^4 y_n + \dots$$

3. Find Solution using Newton's Backward Difference formula

x	f(x)
0	1
1	0
2	1
3	10

Find value $f(4) = ?$

Solution:

The value of table for x and y

x	0	1	2	3
y	1	0	1	10

Newton's backward difference interpolation method to find solution

Newton's backward difference table is

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$
0	1			
		-1		
1	0		2	
		1		6
2	1		8	
		9		
3	10			

The value of x at you want to find the $f(x)$: $x = 4$

$$h = x_1 - x_0 = 1 - 0 = 1$$

$$p = \frac{x - x_n}{h} = \frac{4 - 3}{1} = 1$$

Newton's backward difference interpolation formula is

$$y(x) = y_n + p\nabla y_n + \frac{p(p+1)}{2!} \cdot \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \cdot \nabla^3 y_n + \frac{p(p+1)(p+2)(p+3)}{4!} \cdot \nabla^4 y_n + \dots$$

$$y(4) = 10 + 1 \times 9 + \frac{1(1+1)}{2} \times 8 + \frac{1(1+1)(1+2)}{6} \times 6$$

$$y(4) = 10 + 9 + 8 + 6$$

$$y(4) = 33$$

Solution of newton's backward interpolation method $y(4) = 33$

(Solved)

4.3 Lagrange's Interpolation Formula

Let $y(x)$ be continuous and differentiable $(n+1)$ times in the interval (a,b) , Given the $(n+1)$ points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$ where the values of x need not necessarily be equally spaced, we wish to find a polynomial of degree n , say $L_n(x)$, such that

$$L_n(x) = \sum_{i=0}^n l_i(x)y_j \dots\dots\dots(1)$$

Where $l_i(x)$ are polynomials in x of degree n . Since $l_n(x_j) = y_j$ for $j = 0, 1, \dots, n$, Equation (1) gives

$$\left. \begin{array}{l} l_i(x_j) = 0 \quad \text{if } i \neq j \\ l_j(x_j) = 1 \quad \text{for all } j \end{array} \right]$$

Hence $l_i(x)$ may be written as

$$l_i(x) = \frac{(x-x_0)(x-x_1)\dots(x-x_{i-1})(x-x_{i+1})\dots(x-x_n)}{(x-x_0)(x-x_1)\dots(x-x_{i-1})(x-x_{i+1})\dots(x-x_n)} \dots\dots\dots(2)$$

Simplified Form for example –

Suppose we are to evaluate a polynomial that is passing through three points $(x_0, y_0), (x_1, y_1)$ and (x_2, y_2) then using the formula (1)

$$\begin{aligned} L_2(x) &= \sum_{i=0}^2 l_i(x)y_j \\ &= l_0(x)y_1 + l_1(x)y_1 + l_2(x)y(2) \end{aligned}$$

$$l_0(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x-x_2)}$$

$$l_1(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}$$

$$l_2(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}$$

Now we can evaluate the polynomial using the given formula.

Example –

1. Find Solution using Lagrange's Interpolation formula

x	f(x)
300	2.4771
304	2.4829
305	2.4843
307	2.4871

find value at x = 301

Solution:

The value of table for x and y

x	300	304	305	307
y	2.4771	2.4829	2.4843	2.4871

Lagrange's Interpolating Polynomial

The value of x at we want to find $P_n(x)$: $x = 301$

Lagrange's formula is,

$$L_3(x) = \sum_{i=0}^3 l_i(x)y_i$$

$$= l_0(x)y_0 + l_1(x)y_1 + l_2(x)y_2 + l_3(x)y_3$$

$$= \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x-x_2)(x_0-x_3)}y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)}y_1$$

$$+ \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)}y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)}y_3$$

$$y(301) = \frac{(-3)(-4)(-6)}{(-4)(-5)(-7)} \times 2.4771 + \frac{(1)(-4)(-6)}{(4)(-1)(-3)} \times 2.4829 + \frac{(1)(-3)(-6)}{(5)(1)(-2)} \times 2.4843$$

$$+ \frac{(1)(-3)(-4)}{(7)(3)(2)} \times 2.4871$$

$$y(301) = -\frac{72}{140} \times 2.4771 + \frac{24}{12} \times 2.4829 + \frac{18}{-10} \times 2.4843 + \frac{12}{42} \times 2.4871$$

$$y(301) = 2.4786$$

(Solved)

4.4 Newton's divided difference Formula

Suppose we are to evaluate a polynomial that is passing through three points (x_0, y_0) , (x_1, y_1) and (x_2, y_2) using the divided difference formula

$$y(x) = y_0 + (x - x_0)f[x_0, x_1] + (x - x_0)(x - x_1)f[x_0, x_1, x_2] + \dots$$

Where, $f[x_0, x_1] = \frac{y_1 - y_0}{x_1 - x_0}$,

$$f[x_1, x_2] = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\text{And } f[x_0, x_1, x_2] = \frac{f[x_0, x_1] - f[x_1, x_2]}{x_0 - x_2}$$

Example 2.

Find Solution using Newton's Divided Difference Interpolation formula

x	$f(x)$
2	0.69315
2.5	0.91629
3	1.09861

Evaluate $f(x)$ when $x = 2.7$.

Solution:

Newton's divided difference table is

x	y	1st order	2nd order
2	0.69315		
		0.44628	
2.5	0.91629		-0.08164
		0.36464	
3	1.09861		

$$\frac{0.91629 - 0.69315}{2.5 - 2} = 0.44628$$

$$\frac{1.09861 - 0.91629}{3 - 2.5} = 0.36464$$

$$\frac{0.36464 - 0.44628}{3 - 2} = -0.08164$$

The value of x at we want to find the $f(x)$: $x = 2.7$

Newton's divided difference interpolation formula is

$$y(x) = y_0 + (x - x_0)f[x_0, x_1] + (x - x_0)(x - x_1)f[x_0, x_1, x_2] + \dots$$

$$y(2.7) = 0.69315 + (2.7 - 2) \times 0.44628 + (2.7 - 2)(2.7 - 2.5) \times -0.08164$$

$$y(2.7) = 0.69315 + (0.7) \times 0.44628 + (0.7)(0.2) \times -0.08164$$

$$y(2.7) = 0.69315 + 0.312396 - 0.01143$$

$$y(2.7) = 0.994116$$

Solution of divided difference interpolation method $y(2.7) = 0.994116$.

(Solved)

Exercises –

(1) State Newton's forward and backward interpolating formula.

(2)

The table gives the distances in nautical miles of the visible horizon for the given heights in feet above the earth's surface: Find the values of y when (i) $x = 218$ ft. (ii) $x = 410$ ft.

x=height	100	150	200	250	300	350	400
y=distance	10.63	13.03	15.04	16.81	18.42	19.90	21.27

(3) Find Newton's forward difference form of interpolating polynomial for the data. Hence interpolate $f(5)$.

x	4	6	8	10
f(x)	19	40	79	142

(4) Estimate the value of $f(1.45)$ from the data given below

x	1.1	1.2	1.3	1.4	1.5
f(x)	1.3357	1.5095	1.6984	1.9043	2.1293

(5) Find the Newton's divided difference form of interpolating polynomial for the data.

x	-4	-1	0	2	5
f(x)	1245	33	5	9	1335

Also approximate $f(1)$ from the polynomial.

(6) From the table of values given below, obtain the value of y when $x = 1$ using

(a) Divided difference interpolation formula.

(b) Lagrange's interpolation formula

x	0	2	3	4
f(x)	-4	6	26	64